# Day 16 – Selection Sort

🔷 Selection Sort – Deep Explanation

💡 Basic Idea

👉 “Har pass me smallest element ko dhundho aur usko correct position par rakh do”

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🧠 Simple Language Trick

👉 Socho tum exam me ho aur roll numbers ko arrange karna hai:

- ✓ Pehle sabse chhota number dhundho
- ✓ Usko first position pe rakho
- ✓ Phir next smallest dhundho
- ✓ Usko second position pe rakho

👉 Ye hi Selection Sort hai 😄

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🔁 Step-by-Step Dry Run

Example

arr = [64, 25, 12, 22, 11]

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🔹 Pass 1

👉 Find minimum from index 0 to end

Min = 11 (index 4)

Swap with index 0:

[11, 25, 12, 22, 64]

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🔹 Pass 2

👉 Find minimum from index 1 onward

Min = 12 (index 2)

Swap:

[11, 12, 25, 22, 64]

- Pass 3

Min = 22

[11, 12, 22, 25, 64]

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- Pass 4

Already sorted

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🔥 Final Output

11 ,12 ,22 ,25, 64

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2️⃣ Java Code (Selection Sort)

```java
package Day_16;

public class SelectionSort {

    public static void selectionsort(int[] arr){

        int n = arr.length;

        for(int i = 0; i < n - 1; i++){

            int minIndex = i;

            // minimum find karna
            for(int j = i + 1; j < n; j++){

                if(arr[j] < arr[minIndex]){
                    minIndex = j;
                }
            }

            // swap
            int temp = arr[i];
            arr[i] = arr[minIndex];
            arr[minIndex] = temp;
        }
    }

    public static void main(String[] args) {

        int[] arr = {29,10,14,37,13};

        selectionsort(arr);

        for(int num : arr){
            System.out.print(num + " ");
        }
    }
}
```

3️⃣ Important Observations 🔍

✅ Swaps kam hote hain

👉 Har pass me sirf 1 swap

👉 Total swaps = n-1

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❌ Comparisons zyada hote hain

👉 Har element ko compare karna padta hai

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4️⃣ Time Complexity

| Case | Time |
|---|---|
| **Best** | $O(n^2)$ |
| **Worst** | $O(n^2)$ |
| **Average** | $O(n^2)$ |

👉 Bubble sort me best case O(n) ho sakta hai
👉 But Selection sort me always $O(n^2)$ ❌

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5️⃣ Bubble vs Selection (Important 🔥 )

| Feature | Bubble Sort | Selection Sort |
|---|---|---|
| **Swap** | zyada | kam |
| **Best Case** | O(n) | $O(n^2)$ |
| **Stability** | Stable | Not stable |
| **Logic** | adjacent swap | min select |

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6️⃣ Descending Order

👉 Bas condition change karo:

```
if(arr[j] > arr[minIndex])
```

## 7️⃣ Interview Questions

Q1: Selection Sort kab use kare?

👉 Jab swaps minimize karna ho

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Q2: Kya ye stable hai?

👉 ❌ Nahi

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Q3: Kya ye fast hai?

👉 ❌ Large data ke liye slow hai

**Problem 1**

Descending order me likho

```java
package Day_16;

public class SelectionSortDec {
     public static void selectionsort(int[] arr){

        int n = arr.length;

        for(int i = 0; i < n - 1; i++){

            int minIndex = i;

            // minimum find karna
            for(int j = i + 1; j < n; j++){

                if(arr[j] > arr[minIndex]){  /// for Desending order
                    minIndex = j;
                }
            }

            // swap
            int temp = arr[i];
            arr[i] = arr[minIndex];
            arr[minIndex] = temp;
        }
    }

    public static void main(String[] args) {

        int[] arr = {29,10,14,37,13};

        selectionsort(arr);

        for(int num : arr){
            System.out.print(num + " ");
        }
    }
}
```

## Problem 2

Count swaps

```java
package Day_16;

public class SelectionSortCountSwap {
    public static void selectionSort(int[] arr){

        int n = arr.length;
        int swapCount = 0;

        for(int i = 0; i < n-1; i++){

            int minIndex = i;

            for(int j = i+1; j < n; j++){
                if(arr[j] < arr[minIndex]){
                    minIndex = j;
                }
            }

            // swap only if needed
            if(minIndex != i){

                int temp = arr[i];
                arr[i] = arr[minIndex];
                arr[minIndex] = temp;

                swapCount++; // count swap
            }
        }

        System.out.println("Total swaps = " + swapCount);
    }

    public static void main(String[] args){

        int[] arr = {29,10,14,37,13};

        selectionSort(arr);

        for(int num : arr){
            System.out.print(num + " ");
        }
    }

}
```